

Diffraction Grating

HLM UNIVERSITY, GREATER UNIVERSITY PHYSICS LABORATORY MANUAL
PRACTICAL EXPERIMENT INSTRUCTION SHEET

EXPERIMENT TITLE: To determine the wavelength of a given laser source using diffraction grating.

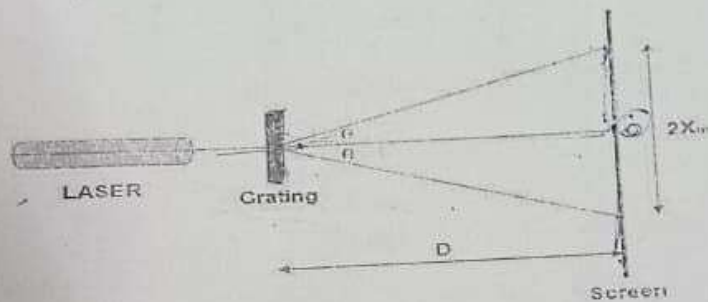
EXPERIMENT NO. 07 DEPT: Physics, School of Sciences ISSUE DATE: 07 July, 2023

LABORATORY: PHYSICS SEMESTER: I, II TOTAL PAGE: 4
LAB (UBS-1008)

AIM: To determine the wavelength of a given laser source using diffraction grating.

APPARATUS: Laser Source, grating of known grating element, optical bench, laser tube holder, screen and meter scale.

DIAGRAM:



FORMULA USED:

Grating element of a given grating can be calculated by using the formula
 $(a+b) \sin \theta = n\lambda$.

Here $(a+b)$ is the grating element, ' n ' is the order of diffraction and θ is angle of diffraction and can be calculated by using the formula

$$\sin \theta = \frac{X_m}{\sqrt{X_m^2 + D^2}}$$

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PROCEDURE:

1. Place the optical bench on a table at approximately 1 to 2 meter away from the screen and mount the grating of known grating element and laser tube on their respective uprights so that the face of the grating is perpendicular to the length of the optical bench and the laser beam falls normally on the grating when the laser power supply is switched on. Adjust the heights of laser tube and grating so that the laser beam passes through the center of the grating. The diffraction spots on the screen as shown in the figure will be observed.
2. Measure the center-to-center distance between 1 order diffraction spots with meter scale, let it be $2X_m$.
3. Measure the distance D from the grating face to screen and calculate the value of λ for the laser source using the formula

$$(a+b) \sin \theta = \lambda; \quad (n=1)$$

If N is the number of lines/inch, then $(a+b)$ is given by

$$(a+b) = 2.54/N$$

$\sin \theta$ is given by-

$$\sin \theta = \frac{X_m}{\sqrt{X_m^2 + D^2}}$$

4. Repeat experiment for three different values of D and calculate the mean value of λ .
5. Now replace the grating of known grating element with the grating of unknown grating element. Repeat the steps 1 to 2 for different values of D and find the grating element of this grating using the formula

$$(a+b) = \lambda / \sin \theta$$

6. Use the value of λ that has been obtained in step (3) and calculate the value $\sin \theta$.

OBSERVATIONS

- (i)
- (ii)

OBSERVATION

Sr. No.	Distance between grating screen (D in cm)
1.	
2.	
3.	

Calculations

Percentage

RESULT:

- 1.
- 2.
- 3.

OBSERVATIONS:

- (i) The number of lines of grating $N=15000$ lines/inch
- (ii) Standard value of wavelength of given laser _____

OBSERVATION TABLE

Sr. No.	Distance between grating & screen (D in cm)	Distance between the first order maxima ($2X_m$ in cm)	X_m (cm)	$\frac{\sin \theta}{\sqrt{m^2 + D^2}}$	$\lambda = (a+b) \cdot \sin \theta (\text{\AA})$	Mean λ ($\text{\AA}$)
1.						
2.						
3.						

Calculations: =

Percentage error =

$$\frac{\lambda_{\text{standard}} - \lambda_{\text{calculated}}}{\lambda_{\text{standard}}} \times 100$$

RESULT:

1. The wavelength of a given laser source is $\lambda = \dots \text{\AA}$
2. Standard wavelength of λ of the given laser source = $6300 \text{\AA}$
3. Percentage Error =

PRECAUTIONS:

1. Never point the laser light towards the eyes.
2. Uprights should be at the same heights.
3. Laser light should be incident on the grating normally at the centre.
4. Position of the laser source should be fixed.

VIVA QUESTION

1. What is diffraction grating?
2. What is grating element?
3. What is the full form of laser?
4. Why does laser is more monochromatic than ordinary source of light?
5. What is the required condition for lasing action?

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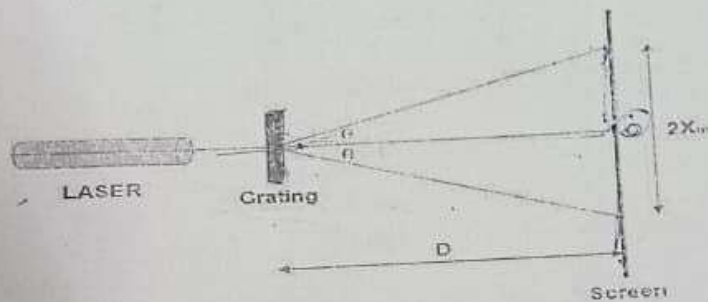
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